

1. High-Level Transaction Lifecycle

Reservation Creation → Availability Verification → Confirmation

→ Checkout (Transaction Created) → Active Rental Period

→ Return (Status Update on same Transaction)

→ Billing & Payment → [Conditional: Damage/Penalty Assessment]

→ Transaction Closure & Reporting

Stage	Type	Description
1. Reservation Creation	Creation	Customer selects equipment + dates, submits request
2. Availability Verification	Processing	Staff cross-checks the item is actually free for those dates
3. Confirmation	Processing	Staff approves, rejects, or proposes an alternate
4. Checkout	Fulfillment (start)	One form. Staff creates the Rental Transaction — this single action <i>is</i> the pickup confirmation (sets CheckoutDate)
5. Active Rental Period	In-progress	No action needed — the system identifies "active" as any transaction where ReturnDate is still empty
6. Return	Fulfillment (end)	One form. Staff updates that same transaction — sets ReturnDate and condition notes; no separate return record
7. Billing & Payment	Completion	Final invoice generated and settled
8. Damage/Penalty Assessment	Completion (conditional)	Only triggers if condition notes flag damage on return
9. Transaction Closure & Reporting	Completion/Archival	Status field updated to Completed; feeds into Reports

2. Role-Based Flow & Interactions

Staff

Entry point: Operations dashboard — "Today's Reservations / Pickups / Returns" queue

Stage	Trigger / Input	Permission	Modules Touched
Verification	New request appears in queue	Read/list	Reservation Mgmt (read)
Confirmation	Checks calendar, approves/rejects/proposes alternate	Update	Equipment Availability (read/update), Reservation Mgmt (update)
Checkout	Customer arrives; Staff fills one "New Transaction" form	Create	Rental Transactions (create — sets CheckoutDate), Equipment Mgmt (status update), Equipment Availability (update)
Active Rental	Monitors due dates via a filtered view (ReturnDate = null)	Read	Rental Transactions (read, filtered)
Return	Customer returns item; Staff opens the <i>same</i> transaction and fills one "Process Return" form	Update	Rental Transactions (update — sets ReturnDate, ConditionNotes, Status), Equipment Mgmt (update), Damage & Penalty Tracking (create if applicable), Equipment Availability (update)
Billing	Finalizes invoice, processes payment	Create/update	Billing (create/update), Rental Transactions (update status)
Damage/Penalty (conditional)	Applies standard fee OR escalates if beyond schedule	Create/escalate	Damage & Penalty Tracking (create, escalate to Admin)
Closure	Updates Status field to Completed on the same transaction	Update	Rental Transactions (update), Maintenance Mgmt (create if needed)

Admin

Entry point: System-wide dashboard — revenue, active rentals, pending escalations

Stage	Trigger / Input	Permission	Modules Touched
Stages 1-6	Generally passive — oversight only, unless correcting an error	Read all + override	All modules (read), Rental Transactions (override, rare)
Damage/Penalty (conditional)	Receives escalation for penalty beyond standard fee schedule	Approve/adjust/waive	Damage & Penalty Tracking (approve/override)
Closure & Reporting	Pulls system-wide reports, may adjust fee/policy going forward	Full read + configure	Reports (full access), System Settings (configure)

3. Role Interaction Matrix (Handoffs)

Stage	Customer	→	Staff	→	Admin
1. Request	Submits reservation	→	—		—
2-3. Verify/Confirm	(waits)		Verifies & confirms/rejects		(only if unresolved conflict)
4. Checkout	Picks up equipment (physical)	→	Creates transaction — 1 form		—
5. Active Rental	Holds equipment		Monitors via filtered view		—
6. Return	Returns equipment (physical)	→	Updates <i>same</i> transaction — 1 form		—
7. Billing	Views & pays invoice	←	Finalizes invoice		—
8. Penalty (conditional)	Views/disputes charge	←	Applies standard fee	→	Approves/waives if escalated
9. Closure	Views closed record	←	Updates status field	→	Reviews aggregate reports